SIZANOBUHLE SPHIWOKUHLE ZULU

Computer Science and Physics | Machine Learning & Data Science
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BACKGROUND

I am a final-year Computer Science and Physics student at the University of Zululand, graduating in December 2026, with practical experience training, evaluating and deploying machine learning models. I fine-tuned a MobileNetV2 CNN to 98.07% validation accuracy across 28 classes, added out-of-distribution detection using feature-space embeddings, and served the model through a live FastAPI production API. My physics background gives me a strong foundation in mathematical modelling and data analysis. Placed 3rd nationally in the TCS Sustainathon South Africa 2026 out of approximately 1,200 entries. Available from January 2027 and open to relocation.

EDUCATION

Bachelor of Science in Computer Science and Physics

University of Zululand, Empangeni, South Africa — 2023 – Expected 2026

- Member of Computer Science Society
- **Relevant Coursework:** Machine Learning, Software Engineering, Database Systems, Data Structures, Distributed Systems, Nuclear Physics, Solid State Physics

CERTIFICATIONS & ACHIEVEMENTS

3rd Place — TCS Sustainathon South Africa 2026

Tata Consultancy Services — September 2026

- Placed 3rd nationally out of approximately 1,200 entries with Khusela, an AI system addressing post-harvest food loss. Selected as a top-10 finalist and presented to a panel of industry judges in Sandton, Johannesburg.

MuleSoft 4.X Complete Guide — Hands On Projects

Udemy — June 2026 | 48.5 hours | Certificate No. UC-a83e71be-1bed-4789-bbe0-9e2cc4df32d9

- Comprehensive hands-on training in MuleSoft Anypoint Platform, covering API-led connectivity, RAML specification, DataWeave transformation, and flow development in Anypoint Studio.

EXPERIENCE

University of Zululand Chatbot | Backend Developer

Empangeni, South Africa — 2024

- Developed a Python-based chatbot using FastAPI and PyTorch, deployed to Azure Functions with 99% uptime.
- Followed OWASP security practices and automated testing through CI/CD pipelines.
- Also contributed to the frontend development with styling using CSS.

University of Zululand | Computer Literacy Tutor

Empangeni, South Africa — Feb 2025 – Nov 2025

- Helped students with Office Suite (Microsoft) apps (Word, Excel, PowerPoint).
- Also helped with Computer Lab experiments using languages like Java and Python.

PROJECTS

Khusela — AI Post-Harvest Produce System | Machine Learning & Backend Developer

GreenByte Solutions — 2026 | 3rd Place, TCS Sustainathon South Africa 2026 | Live: khusela-frontend.vercel.app

- An AI system that grades harvested produce from a phone photograph, predicts its remaining shelf-life, and routes at-risk food to the best destination before it spoils — addressing post-harvest food loss in South Africa.
- Trained and fine-tuned a MobileNetV2 convolutional neural network using two-phase transfer learning across 29,317 images and 28 classes, achieving 98.07% validation accuracy.
- Implemented out-of-distribution detection using feature-space embeddings to reject non-produce images that confidence scoring alone could not catch.
- Built the backend in FastAPI — 27 REST endpoints covering JWT authentication, image handling, a decision engine, and multilingual support in English, isiZulu and isiXhosa.
- Deployed to production and currently live: API on Render, React frontend on Vercel, both served over HTTPS.

PGMS — Perishable Goods Management System | Group Leader & Backend Developer

University of Zululand — 2026 | Final-Year Capstone Project (4CPS322) | Theme: Digital Solutions for Community Empowerment

- Leading a six-person team to build an AI decision-support system that tells South African farmers where, and at what price, to sell perishable produce based on its predicted shelf-life — turning image-based grading into actionable selling decisions.

- Aligned grading with SA regulation (Class 1/2/3, Agricultural Product Standards Act 119 of 1990) and researched Q10 temperature effects on shelf-life in a 15-page research briefing.
- Resolved a model contradiction that returned “Healthy” and “Expired” for the same image by constraining shelf-life predictions to the detected produce type with a confidence-based override; fixed cross-version Keras model loading.
- Developing the FastAPI backend (SQLModel, SQLAlchemy, SQLite) with a province and market-destination data model covering all nine provinces and KZN fresh produce markets and food banks.
- Authored a 32-page IEEE-830 Software Requirements Specification (11 functional requirements, 9 database entities) with UML use-case, activity and system-environment diagrams.

TradeBridge | MuleSoft Integration Developer
Portfolio Project — 2025

- Designed a three-layer API-led connectivity architecture (Experience, Process, System) on MuleSoft Anypoint Platform, using RAML specifications and DataWeave transformations.

TECHNICAL SKILLS

- **Machine Learning & Neural Networks:** Training and fine-tuning convolutional neural networks with PyTorch and torchvision, including transfer learning, layer unfreezing, image classification, model calibration, and out-of-distribution detection using feature-space embeddings.
- **AI-Assisted Development:** Working with agentic AI coding tools in VS Code — directing them across a codebase to plan, refactor and test, reviewing generated code critically, and setting clear constraints so output is verifiable rather than taken on trust.
- **API Design & Integration:** Creating RESTful APIs, handling authentication (JWT, bcrypt), multipart file upload, CORS configuration, and integrating third-party services (Google Cloud Logging, PayFast).
- **Cloud Services & DevOps:** Deploying production services to Render and Vercel with environment configuration, persistent storage and automatic builds from Git; Azure Functions; Google Cloud service accounts and logging; Docker.
- **Database Management:** Designing and managing relational databases (SQLite, SQLAlchemy, Django ORM), including schema design, migrations, and updates.
- **Full-Stack Web Development:** Building and maintaining both frontend (React, TypeScript, Vite) and backend (Django, Django REST Framework, FastAPI) components, including responsive mobile-first layouts, Leaflet mapping, and browser camera and geolocation APIs.
- **Testing & Debugging:** Writing and running unit tests, troubleshooting frontend and backend issues, and using shell commands for database operations.
- **Version Control:** Managing codebase with Git, handling branches, and collaborating effectively in a team.
- **Enterprise Integration (MuleSoft):** API-led connectivity across Experience, Process and System layers; RAML API specification and design; DataWeave transformation; flow development in Anypoint Studio; deployment via Anypoint Exchange and API Manager. Completed 48.5 hours of hands-on training (Udemy, 2026).
- **Payment Systems Integration:** Implementing secure payment flows, digital wallet features, and transaction handling.
- **Event-Driven Architecture:** Triggering and handling events for notifications and logging.

SOFTWARE ENGINEERING SKILLS

- **Project Planning & Documentation:** Creating implementation checklists, API contracts, project documentation, and quick reference guides.
- **UI/UX Design:** Enhancing usability with form validation, input restrictions, improved dashboards, and responsive layouts across devices.
- **Problem Solving:** Diagnosing root causes rather than symptoms — including identifying why a fine-tuned model misclassified out-of-domain inputs, and engineering around it.
- **Collaboration:** Communicating requirements, writing interface specifications so frontend and backend work can proceed in parallel, and integrating feedback from stakeholders.

PROFESSIONAL SKILLS

- **Communication:** Writing clear documentation, commit messages, and explaining technical changes.
- **Adaptability:** Quickly learning new frameworks and technologies based on project needs.
- **Initiative:** Proactively identifying issues, suggesting improvements, and implementing new features.

TECHNICAL PROFICIENCIES

- **Languages:** Python, JavaScript, TypeScript, Java, SQL
- **Frameworks & Libraries:** PyTorch, torchvision, Keras, FastAPI, Django, Django REST Framework, React.js, Vite, SQLAlchemy, SQLModel, Leaflet
- **Databases:** SQLite, MongoDB, PostgreSQL
- **DevOps & Tools:** Git, GitHub Actions, Docker, VS Code, MuleSoft Anypoint Platform, Google Colab, Google Cloud Logging
- **Cloud Platforms:** Azure, Google Cloud, Render, Vercel